

Aftershock

Earth science · statics & materials

Kestrel Bay three days after, where the fault broke the surface and you can walk the scarp.

45	22	3.8	2/10	6	28 · 17 · 0
STOPS	CHOICE (49%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

The worst curriculum delivery in the set: 2 of 10 equations computed. Four of the eight gaps sit on instruments and may be Phase 0 phantoms, but four do not, and two equations — a building’s natural period and base shear — are mentioned **nowhere**, with base shear also listed as an absent concept. Twenty-two CHOICE stops, seven select-only concepts, and **zero OPERATE stops** — the only game in the set with none. The place, by contrast, is one of the strongest: granite bench above, liquefied fill below, and a 1.8 m fault scarp you walk between them.

NOW The mix as it ships

CHOICE 22	BALLPARK 4	SEQUENCE 3	CASEBOOK 3	PROTOCOL 2	TRIAGE 2	VALUE 2	ATTEST 2	ALLOCATE 1	CHAIN 1	SCIENCETANK 1
TRACE 1	DELEGATE 1									

CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$M_0 = \mu \cdot A \cdot D$	seismic moment, and what makes a big earthquake big	5 · CHOICE 34 · TRACE	verify first
$M_w = \frac{2}{3} \log_{10} M_0 - 6.06$	moment magnitude, and why the scale is logarithmic	5 · CHOICE 34 · TRACE	verify first
$T \approx 0.1 N$	a building's natural period from its storeys	– no stop at all	real gap
$V = C_s \cdot W$	base shear — the sideways force it is designed for	– no stop at all	real gap
$\sigma = F / A, \epsilon = \Delta L / L$	stress and strain, and the yield point between them	2 · PROTOCOL 10 · CHOICE 17 · CHOICE 22 · CHOICE 23 · CHOICE	real gap
$\sigma' = \sigma - u$	effective stress, and how liquefaction happens	2 · PROTOCOL 10 · CHOICE 11 · CHOICE 26 · CHOICE 38 · CHOICE	real gap
$n(t) = K / (c + t)^p$	aftershock rate, falling as a power law	1 · CHOICE 6 · SEQUENCE 8 · CHOICE 10 · CHOICE 17 · CHOICE	verify first
$A_{\text{soft}} / A_{\text{rock}}$	site amplification — the same wave on two grounds	1 · CHOICE 2 · PROTOCOL 7 · CASEBOOK 11 · CHOICE 31 · CHOICE	real gap

3 of 8 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 5 are unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM	CONCEPT	STOPS
8	Natural period, and resonance between building and ground		1
11	Ductility: bending without breaking		23
14	Unreinforced masonry, and why it fails outward		31, 33, 35
17	Lateral spreading toward a free face		26
22	Rapid assessment against detailed evaluation		7, 8, 9, 15
27	Recurrence intervals and what "the big one" means		33, 43

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects. The syllabus also lists 2 concepts no stop touches at all — **Intensity against magnitude — one event, many shakings; Base shear — the horizontal force a building is designed for** — which is expected, and is a different conversation from this one.

FUN What this subject and this place can carry

TALLY STRONG

stop 19 or 20 · HAZ / SEIS

Aftershocks accumulate and the rate falls as a power law. Omori's law is mentioned at **fifteen** stops and computed at none, and "a rate is not a schedule" is currently a CHOICE. TALLY is the format whose subject is precisely that an early number moves because there is not enough behind it.

SERVES $n(t) = K/(c+t)^p$ — the most-mentioned uncomputed equation in the repo

NEEDS an early batch that flatters the rate, and a pass mark that stays unprinted

PROBE STRONG

stop 26 or 40 · GEO

Walk from the granite bench across the scarp onto the fill, taking a reading at each station, and name where the ground changes. Site amplification is mentioned at thirteen stops and computed at none, and the place was built to be walked exactly this way.

SERVES $A_{\text{soft}} / A_{\text{rock}}$ — and the scarp stops being scenery

NEEDS stations along the real traverse; one honest break that is not the worst number

SWEEP STRONG

stop 1 or 11 · SEIS / STRUCT

Building period against ground period, and the peak is where they meet. Concept 8 — natural period, and resonance between building and ground — is select-only on one stop, and it is the single most consequential idea in earthquake engineering.

SERVES concept 8, and $T \approx 0.1 N$, which is mentioned nowhere at all

NEEDS a findable peak, unmarked

SPOT STRONG

stop 8 or 9 · SAFE / HAZ

A placarding criterion replaced mid-round with nothing announcing it. "Green does not mean safe" is already the game's own sentence; a rapid-assessment crew whose criterion is revised on day three is what actually happened in every event this game is drawn from.

SERVES concept 22 — rapid assessment against detailed evaluation, currently select-only

RISK score only the discriminating items

TRIAL STRONG

the bay

An assessment round driven across the scarp: which buildings first, and the walk between them is the geology. Order is the answer — the school before the parade building, because four hundred children arrive on Monday.

SERVES concept 22 and triage order

RISK world work, and the most expensive item in this plan

HOLD MEDIUM

STRUCT · shoring

Hold a shoring jack's load inside a band while the structure settles and aftershocks arrive. Real, and the physics is statics rather than the dynamics the syllabus is short of.

BELT DEFER

Sorting placards at speed would teach the criterion and reward speed on a subject where speed killed people. Defer on tone.

LOB REJECT

Reject, and not only on subject.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: curriculumDelivery.mjs aftershock --snapshot before.json , then --against before.json .

Minimum honest 3 STOPS · ONE SITTING

19	d7	HAZ	BALLPARK → TALLY	The rate, falling the rate, falling — accumulated so Omori is computed
26	d9	GEO	CHOICE → PROBE	Soft ground, three days on soft ground three days on, walked station by station
11	d4	STRUCT	CHOICE → SWEEP	What eight degrees does what eight degrees does — as a resonance curve

CHOICE 22 → 20 (44%) · moves 28·17·0 → 26·16·3 · effective formats 3.8 → 4.5 · 3 new operate stops

Balanced 6 STOPS · THE RECOMMENDATION

19	d7	HAZ	BALLPARK → TALLY	The rate, falling
26	d9	GEO	CHOICE → PROBE	Soft ground, three days on
11	d4	STRUCT	CHOICE → SWEEP	What eight degrees does
22	d8	MAT	CHOICE → BALLPARK	A number, with a method attached a number with a method attached — stress and strain, computed
8	d3	SAFE	CHOICE → SPOT	Green does not mean safe green does not mean safe, under a criterion that changes
31	d11	STRUCT	CHOICE → CHAIN	Walls that fall outward walls that fall outward — name the link that governs

CHOICE 22 → 17 (38%) · moves 28·17·0 → 23·18·4 · effective formats 3.8 → 5.8 · 4 new operate stops, 1 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

19	d7	HAZ	BALLPARK → TALLY	The rate, falling
26	d9	GEO	CHOICE → PROBE	Soft ground, three days on
11	d4	STRUCT	CHOICE → SWEEP	What eight degrees does
22	d8	MAT	CHOICE → BALLPARK	A number, with a method attached
8	d3	SAFE	CHOICE → SPOT	Green does not mean safe
31	d11	STRUCT	CHOICE → CHAIN	Walls that fall outward
39	d13	STRUCT	CHOICE → CLOUD	A smaller shake against a weaker building a smaller shake against a weaker building, as a band against a limit
23	d8	STRUCT	CHOICE → HOLD	Which way the cracks run shoring held while the aftershocks arrive
9	d3	HAZ	TRIAGE → TRIAL	The ones worth a second visit the assessment round, driven across the scarp

CHOICE **22 → 15** (33%) · moves **28·17·0 → 20·19·6** · effective formats **3.8 → 7.1** · **6** new operate stops, 3 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. **2.** Snapshot. **3.** Edit the book, never the generated content — tools/import-book.mjs books/aftershock.yml aftershock --verify. **4.** npm run traps: every instrument's trap must fire. **5.** npm run check aftershock and npm run drive aftershock, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. **6.** --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from themes/aftershock/content/ via engine/dev/curriculumDelivery.mjs; concept and equation lists authored in tools/syllabus.js. Format verdicts and slates are editorial. Run it with THEME=aftershock npm run dev.